Biological and climatic constraints drive reproductive synchrony

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Abstract

Synchronous 'boom-and-bust' reproduction—or masting—is common across tree species globally and hypothesized to increase regeneration. Decades of research have identified climate as a potential driver of the timing of masting, raising concerns anthropogenic climate change may disrupt it. Yet exactly how trees reproduce synchronously—a critical component of masting—is harder to explain. Here, using long-term seed production data from individual beech trees across England, we scale up tree-level predictions to population-level dynamics, to show how climate and biological constraints together drive reproductive synchrony. Our approach integrates climatic cues with a reproductive constraint within each tree: overlap between floral bud initiation and fruit maturation that drives a trade-off between the two processes across years. Our results support the idea of masting as a synchronous 'boom-and-bust' cycle, with 87% of trees within each population sharing the same reproductive state in any given year. Synchrony between populations did not decline significantly over the last decade despite marked warming (84% before 2015 versus 83% after). Warm summers increased the probability that trees transitioned into a high reproductive state, but individual-level reproductive constraints prevented runaway increases in masting frequency, even under forecasts of extreme warming. We found synchrony was driven mainly by cold summers, which pushed most trees into a shared low reproductive state, providing the conditions for synchronous reproduction when a warm summer followed. Our results suggest the current interplay of individual-level constraints and population-level climatic cues will lead to continued synchrony with warming in the immediate future, and highlight the need for more tree-level data and mechanistic models for accurate forecasts.

Introduction

The acceleration of climate change may cause abrupt ecological effects worldwide (Trisos *et al.*, 2020). Increasing extreme conditions could disrupt key ecological processes, and drive ecosystems toward critical transitions (Wernberg *et al.*, 2016). Across biomes, recent work has suggested major shifts in pests and pathogens, while the local extirpation of rare species and increase in invaders across taxa has driven biotic homogenization (Carlson *et al.*, 2025; Dornelas *et al.*, 2014; Daru *et al.*, 2021). Forests in particular appear to be shifting, with lagging elevational and latitudinal range expansions, alongside increases in adult tree mortality (McDowell *et al.*, 2020). This has driven growing concerns about the future of forests to act as carbon sinks globally (Albrich *et al.*, 2020;

Forzieri *et al.*, 2022; McDowell *et al.*, 2020), and thus growing pressure to understand the mechanisms that underlie forest regeneration (Foest *et al.*, 2024; Hacket-Pain *et al.*, 2025; Chakraborty *et al.*, 2024; Foest *et al.*, 2026).

Regeneration in many temperate and tropical forests depends on the widely observed phenomenon of masting. Masting—defined by 'boom and bust' cycles of reproduction that are synchronous across most individuals of a population (Janzen, 1978)—occurs across many trees species and is hypothesized to drive fitness benefits during reproduction and the early stages of establishment (Kelly, 1994; Bogdziewicz *et al.*, 2024). The most commonly invoked fitness benefit, 'predator satiation,' hypothesizes that years of high seed production overwhelm seed predators and thus allows a higher proportion of seeds and seedlings to escape predation and establish (Janzen, 1971; Kelly, 1994). Fitness benefits of masting may also operate earlier in reproduction, by increasing pollen exchange and genetic outcrossing (Carlson *et al.*, 2014; Bontrager & Angert, 2019). While these different hypotheses vary in their exact mechanism and life stage, they all imply that the benefits of masting depend on an economy of scale: the benefits increase as more trees within a population (or location) reproduce synchronously.

Despite the importance of synchrony to yielding fitness benefits, how synchronous masting is and how it occurs remain poorly understood. Most hypotheses for how synchrony emerges focus on shared environmental cues at the species-level (Kelly & Sork, 2002; Bogdziewicz *et al.*, 2024). Among many potential cues, an abundance of work suggests that ideal climatic conditions during bud formation may lead to masting (Nussbaumer *et al.*, 2018; Journé *et al.*, 2025). In temperate systems, ideal conditions include sufficiently warm temperatures during the growing season before masting—when reproductive buds generally develop—alongside mild spring conditions that do not cause tissue loss after buds form (i.e., no late spring frosts, Vacchiano *et al.*, 2017; Gallego Zamorano *et al.*, 2016). Climate change, however, is rapidly shifting temperatures across the spring and summer, including the prevalence of spring frosts (Zohner *et al.*, 2020; Chamberlain *et al.*, 2021), and thus could disrupt masting dynamics. Such changes would have major implications for forest resilience (Bogdziewicz *et al.*, 2023; Foest *et al.*, 2024), but forecasting them requires better understanding how these climatic cues affect reproductive biology, from buds to fruits.

Most temperate woody species develop reproductive buds through a multi-year process. In the common two-year reproductive cycle (Fig. 1A-B), bud formation begins the year before flowering and seedset—potentially explaining why summer temperatures from the previous year appear to trigger masting (Geber *et al.*, 1997; Wilkie *et al.*, 2008). The following year some of these buds may be lost to spring frost—potentially reducing the number of seeds in what otherwise would have been a masting year. This two-year cycle, however, also introduces a constraint in most species (Fig. 1C). Because floral buds for the following year develop simultaneously with current-year fruit (Geber *et al.*, 1997), a large fruit crop can depress bud initiation (through hormonal inhibition and within-plant resource competition, Monselise & Goldschmidt, 1982; Bangerth, 2009; Milyaev *et al.*, 2022). The result is a physiological constraint on flower and fruit development that is often invoked to explain why many woody species show alternate bearing—with a large crop production year ('on-year') often followed by one or several years of low production ('off-years', Monselise &

Goldschmidt, 1982; Wilkie *et al.*, 2008). Effectively, these constraints can be conceptualized as producing latent reproductive states (low or high) that shape seed production in the following year (observed as off- and on-years in crops).

These reproductive constraints and states may shape how climate—and climate change—impacts seed production. Ideal climatic conditions should trigger higher investment in floral buds only for trees not already in a high reproductive state (Fig. 1D). Warm summers would therefore not affect all trees equally: trees in a high reproductive state should be largely unaffected, while those in a low reproductive state should become more likely to transition to a high state. Once a reproductive state has been established, spring frost may matter, with damage from frosts most likely to reduce potential seed production in trees in a high state. Together, these individual-level reproductive dynamics could scale up to generate population-level synchrony, with masting emerging through a combination of climatic variability and reproductive constraints.

Here we investigate how climate affects masting by explicitly modeling how climatic cues interact with individual reproductive constraints to generate population-level synchrony. Using seed count data collected from individual trees across populations of beech forests of England since 1980, we first test whether individual trees show evidence for alternative reproductive states. We then examine how synchrony arises at the individual level and scales up to populations. Finally, we leverage these results to forecast how future warming may affect both tree-level seed production and population-level synchrony.

Results and discussion

Following the paradigm that masting is defined by of 'boom and bust' reproduction, our model identified high and low reproductive years as alternative latent states (Fig. 2). In high ('boom') reproductive years, the model estimates 168 seeds counted on the ground beneath an individual tree on average (90% uncertainty range: 163 – 173), while in low ('bust') reproductive years they produced 52 seeds on average (48 – 56 seeds per tree). Seed number per tree in low and high reproductive years, however, overlapped (Fig. 2A), with the difference between these two states thus occurring mostly in the opposite extremes: years in the high reproduction latent state included very high seed counts, which would almost never be possible in the low reproduction state, while years in the low reproduction state were characterized by extremely low counts, including a high probability of no observed seeds at all (Fig. 2A, Supp. Fig. S1).

Transitioning between reproductive states (high or low) depended on a tree's state in the previous year (Fig. 2B). In a year with average climatic conditions (mean summer temperatures of 19.7 ℃), a tree in a low reproductive state was most likely to transition to a high reproductive state the next year, with a probability of 64% (61 – 68%), and a 1.8 times lower probability (36%, 32 – 39%) of staying in a low reproductive state (Fig. 2B). Under the same average climate, a tree in a high reproductive state was almost certain to transition to a low reproductive state, with a probability of 99% (97 – 100%), and very unlikely to remain in the same state (1%, 0 – 3%). Trees thus commonly shift into the opposite reproductive state each year, but the low reproductive state

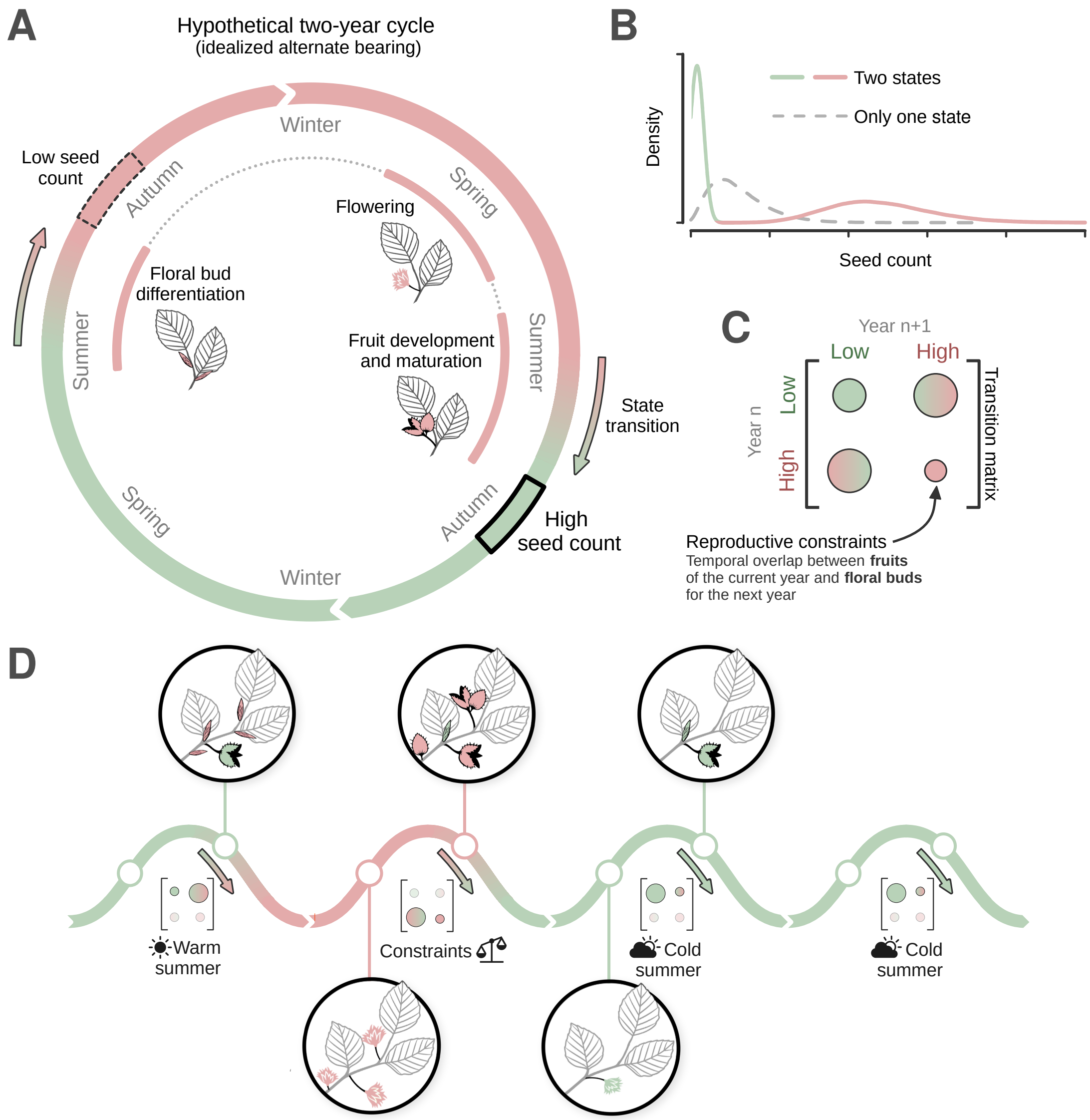

Figure 1: **A two-state model captures the interplay of reproductive constraints and climatic cues. (A)** A large investment in floral buds in summer (initiation and differentiation) leads to a large flowering the following spring (high reproductive state shown in red, transitions shown with arrows). Under favorable conditions, many fruits then develop and mature, producing a high seed count in autumn. Investment in the fruits reduces the investment in floral buds, leading to a low seed count the year after (low reproductive state shown in green). **(B)** High (red) and low (green) reproductive states lead to a different distribution of seed counts, respectively, which a single-state model (gray) would not explain. **(C)** Transitions between low and high reproductive states are governed by a matrix of transition probabilities. The probability of staying in a high state is likely low, reflecting the reproductive constraint: the temporal overlap between maturing fruit in the current year and floral bud formation for the next year. **(D)** On top of this constraint, climate, and summer temperature in particular, influence the transition probabilities. A warm summer can push a tree into a high reproductive state, but the tree is unlikely to stay in a high state the next year because of the reproductive constraint.

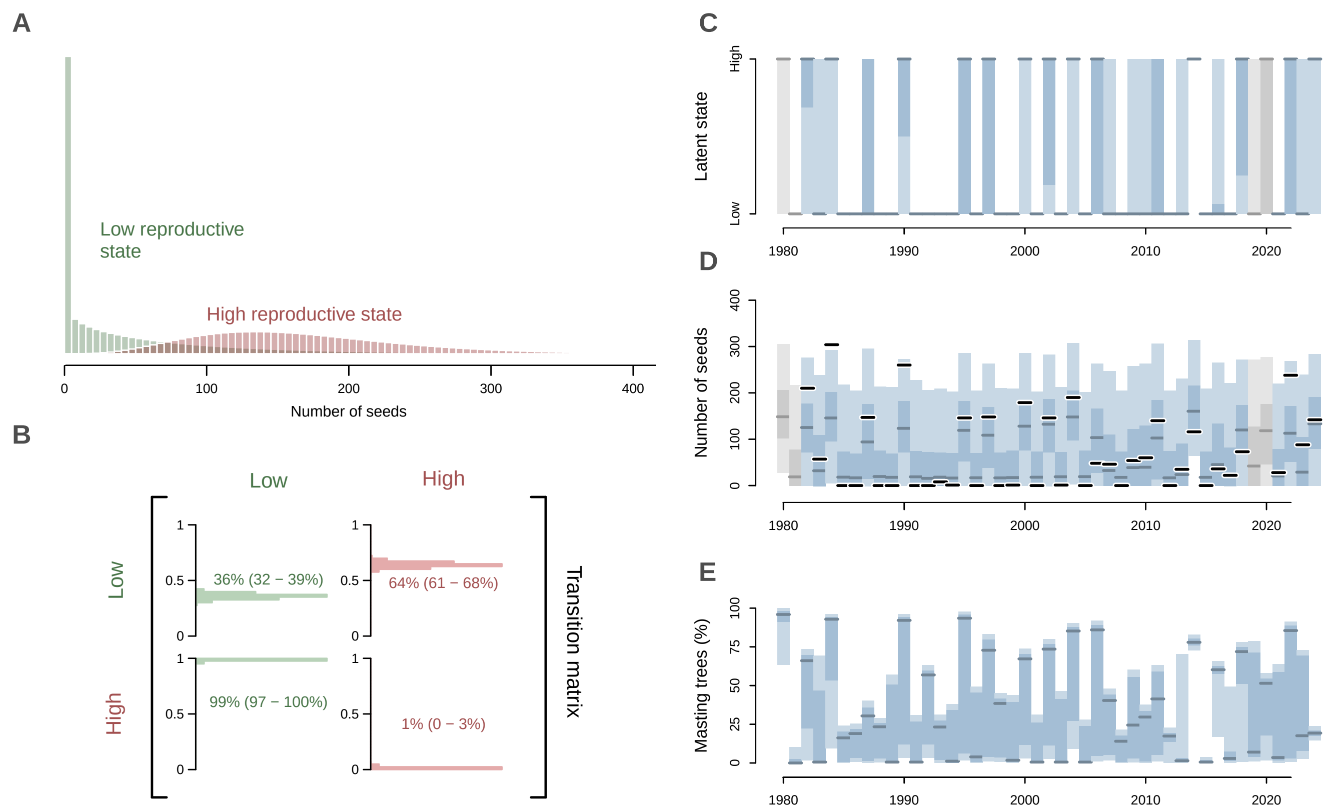

Figure 2: **Distinct tree reproductive states generates variability at the population level.** **(A)** Our two-state model (see Fig. 1) identifies two distinct reproductive latent states (high and low) that correspond to two different levels of seed production. **(B)** Differences in transition probabilities between the two latent states reflect constraints that shape tree-level reproductive cycles. **(C-D)** At the tree level (illustrated here for one arbitrary tree), these alternating latent states over time (**C**) generate low and high seed production years (**D**). **(E)** When added together across all populations, these individual cycles generate variability and synchrony in seed production across years. In **(C)**, **(D)** and **(E)**, we show median model predictions (blue lines) with 50% and 90% uncertainty intervals (shaded areas) overlaid with observed seed counts (black lines) or model predictions for years when observations were missing (gray).

is much more persistent (Fig. 2C-D). Conversely, trees rarely stay in the high reproductive state for more than a single year, supporting the idea of non-mast years being more frequent than mast years (Kelly, 1994).

Climate strongly affected reproductive state transition dynamics. Similar to previous studies (Nussbaumer *et al.*, 2018; Vacchiano *et al.*, 2017; Journé *et al.*, 2024), we found that warm summer temperatures increased the probability that trees in a low reprodutive state transitioned to a high reproductive state (Fig. 3A, Supp. Fig. S2). For example, the transition probability during a warm summer at 22 ℃ was 3.6 times higher than during a 18 ℃ summer (94% vs. 26% respectively, and ranges of 92 – 96% vs. 22 – 30%, Supp. Fig. S3). Additionally, our results support the common hypothesis that spring frosts may limit seed production (Augspurger, 2009; Packham *et al.*, 2012), with seed production for trees in a high reproductive state reduced by 20.1% (13.9 – 26.1%, Fig. 3B) given a three-fold increase in a metric of frost risk (growing-degree days until last frost, Vitasse & Rebetez, 2018), but no evidence that warm springs affected seed production (Supp. Fig. S1). Given the reproductive constraints that prevent beeches and many other woody plants from repetitive years of high fruit (from the competitive overlap of flower bud formation and fruit development, Monselise & Goldschmidt, 1982; Geber *et al.*, 1997), we assumed the transition from a high to a low reproductive state was not affected by climate (we found strong support for this when we relaxed this assumption, Supp. Fig. S4).

Individual tree-level reproductive states (high or low) scaled up to produce population-level seed production that was synchronous, both within and between populations (sites). Within populations, 87% (84 – 89%) of trees shared the same reproductive state on average each year, while synchrony across populations was only slightly lower (the proportion of trees in the same state across all populations was 84%, 81 – 86%). This high level of synchrony across populations could suggest that synchrony occurs across wide spatial scales, as sometimes suggested (Bogdziewicz *et al.*, 2021a; Journé *et al.*, 2024), but given that these populations were not widely separated spatially (Supp. Fig. S5), these results only provide evidence for synchrony over smaller spatial scales, as often suggested (Koenig & Knops, 1998; LaMontagne *et al.*, 2020). Whether this scale is sufficient for fitness benefits depends on the exact mechanism (e.g., predator satiation or pollination efficiency) and the species-level attributes (e.g. predators foraging ranges or wind dispersal distances, which depend on seed shape, size and the local landscape, Davies & MacPherson, 2024). For example, if beech masting in England mainly serves to overwhelm seed predators, such as wood mouse or bank vole (Sachser *et al.*, 2025), then the masting synchrony at the population level that we observe should be sufficient to maintain the regeneration benefits of masting (given that sites are on average 224 km apart, and the two closest sites 1 km apart).

Climatic cues—especially cold summers—appeared critical to synchronizing the reproductive cycle of individual trees to produce the population-level synchrony that often defines masting (Kelly & Sork, 2002). While 36% of trees in a low reproductive state are likely to stay in that state, cold summer temperatures prevent an additional subset from transitioning to a high reproductive state (because cold summer temperatures reduce the probability of transition to a high reproductive state, Fig. 3A). Thus, natural reproductive constraints combined with cold summers can leave most trees in a low reproductive state. This synchrony of low reproduction appears to then drive synchrony of high reproduction when a cold summer is followed by a warm summer; in such years most trees responded consistently to the warm temperatures that promote floral induction, and transitioned together into a high reproductive state (Fig. 3C), generating a mast year at the population scale. This interplay between individual-level constraints and population-level climatic cues provides a physiological mechanism for reproductive synchrony that can persist over multiple years, and offers a simple explanation for previous findings that masting depends on the temperature difference between successive summers (the ΔT model; Kelly *et al.*, 2012).

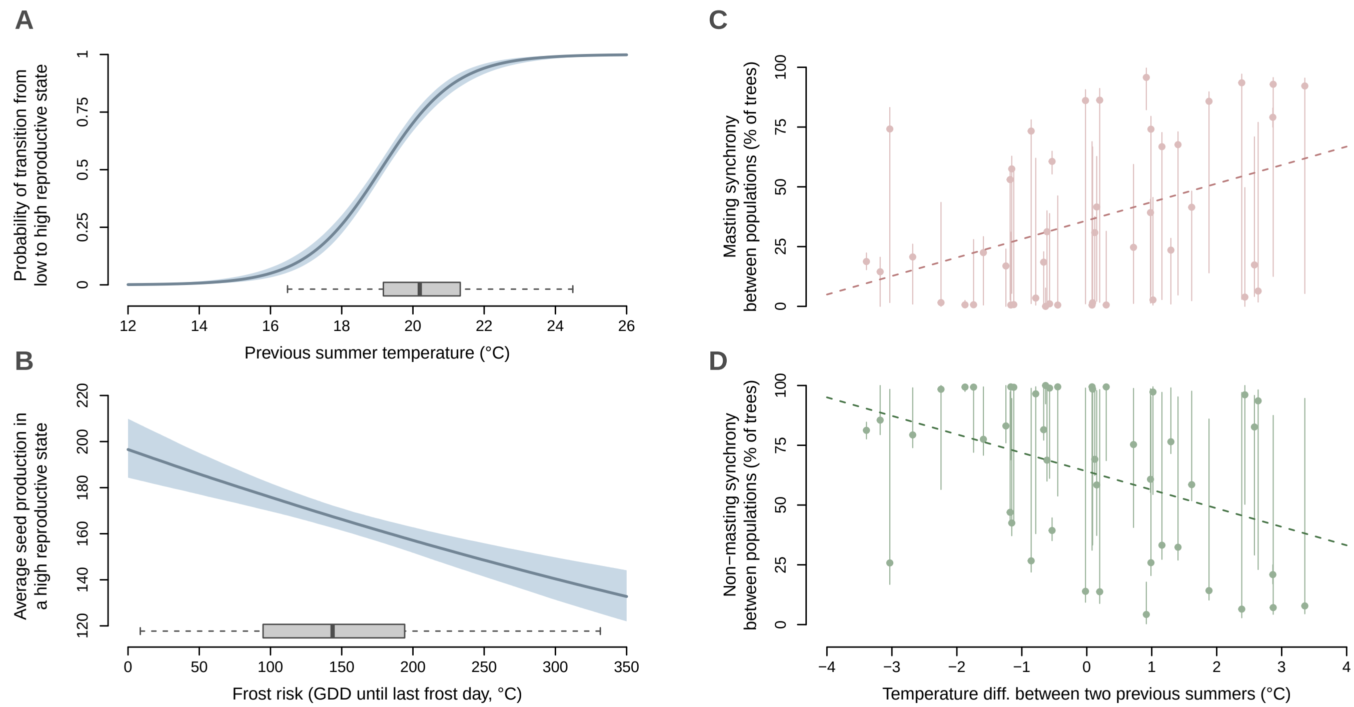

Figure 3: **Climate conditions impact both transition probabilities between reproductive states and seed production. (A)** Average maximum temperature in July and August of the previous year increases the probability of transition from a low to high reproductive state. **(B)** Frost risk (measured as accumulated growing-degree days, GDD, until the last frost day, see Vitasse & Rebetez, 2018) decreases the number of seeds produced when in a high reproductive state. **(C-D)** The interaction of previous-year summer conditions and latent reproductive states (**C**: high reproductive state, **D**: low reproductive state) explain the apparent two-year lag effect of climate on observed synchrony across populations. In **(A)** and **(B)**, the blue line and shaded area respectively represent the median and the 90% uncertainty interval, while the gray boxplots show observed conditions (1980–2022).

Our results appear to support the hypothesis that increasing summer temperatures with climate change could disrupt masting (Bogdziewicz *et al.*, 2021b; Foest *et al.*, 2024; Hacket-Pain *et al.*, 2025). This hypothesis—previously called breakdown—predicts that warm summers will erode both the inter-annual variability of individual tree reproduction and the synchrony of reproduction within populations, i.e. the two components of masting. With warmer summers, individual trees would invest in flowers and fruit in most years, potentially leading to lower tree growth (Koenig &

Knops, 1998; Hacket-Pain *et al.*, 2025). This loss of years of low reproduction would in turn reduce synchrony at the population level, minimizing the fitness benefits of masting (Bogdziewicz *et al.*, 2021b). Because our model captures the reproductive states underlying observed seed production, it allows us to separate these two components—and our results suggest they respond differently to warming.

For inter-annual variability, we find no evidence that warming pushes individual trees towards reproducing in most years. While we do find that warmer summers increase the probability of trees transitioning into a high reproductive state, and therefore increasing summer temperatures would increase the frequency of high seed production years, they do not increase the probability of a tree remaining in that state. Reproductive constraints continue to return trees to a low reproductive state, preventing trees from reproducing heavily in consecutive years. Indeed, we found even with extreme summer warming (7 °C in one century, corresponding to the warmest regional climate model projections, Schumacher *et al.*, 2024), only 52% (40 – 64%) of trees in a population would mast in any given year (Fig. 4). This result holds when we relax reproductive constraints and allow warm summers to increase the probability of trees staying a high-reproductive state (Supp. Fig. S6). Even with this additional effect, we do not observe a breakdown of individual reproduction variability. On the contrary, we found a 'breakdown' because of climate change would require summer warming to have a similar impact on both the transition and persistence into a high reproductive year—an hypothesis not supported by the data used here (Fig. 4).

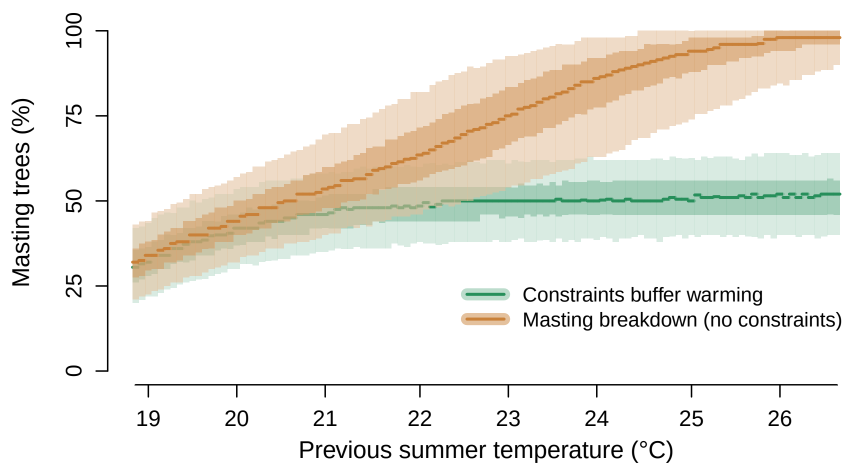

Figure 4: **Reproductive constraints prevent individual trees from reproducing in most years, even under significant summer warming.** Our results suggest trees cannot remain continuously in a high reproductive state with increased warming due to reproductive constraints (green). We could only replicate a potential breakdown of individual reproductive variability with warming through a hypothetical model (not supported by the data here) where we removed reproductive constraints and forced summer warming to increase the probability of persisting into a high reproductive state (brown, see Methods for more details). We show the medians (lines) with 50% and 90% uncertainty intervals (darker and lighter shading respectively) from model predictions.

These results highlight how biological constraints on plant reproduction may limit runaway effects of climate change. For masting, floral bud initiation at the individual tree level is promoted by warm summer temperatures, potentially leading to a higher number of flowers and fruits in the following year (Vacchiano *et al.*, 2017; Bogdziewicz *et al.*, 2024; Journé *et al.*, 2024). In the next summer, however, the development of a larger fruit load inherently imposes a trade-off that limits the tree from initiating a large number of floral buds again (Monselise & Goldschmidt, 1982; Wilkie *et al.*, 2008). These individual-level constraints scale up to the population level by preventing the entire population from producing large seed crops every year—and thus also preventing an unlimited amplification of climate effects on masting. Our results highlight that constraints at the individual level are critical for accurate forecasts of population-scale dynamics (Stearns, 1992). Further this approach can help identify the spatial scale at which changes occur, including possible shifts in synchrony.

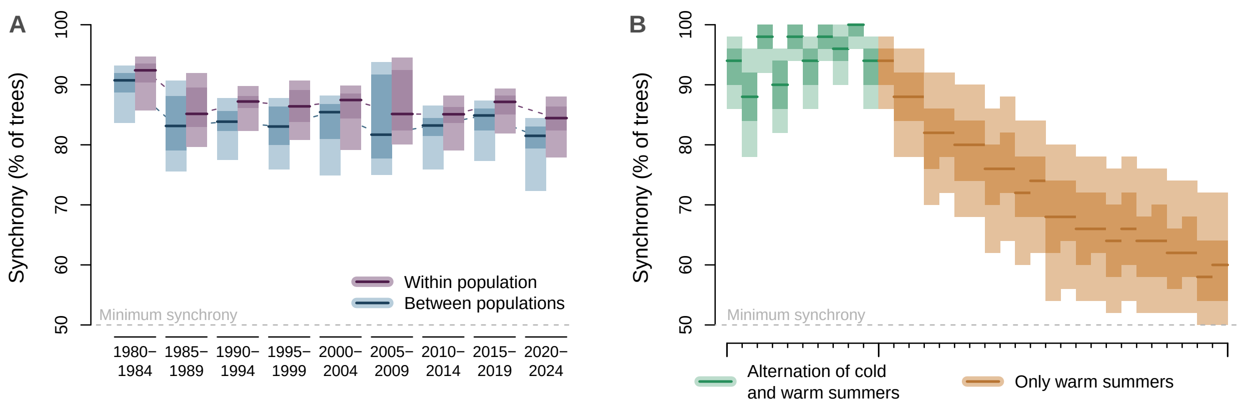

Figure 5: **Reproductive synchrony has remained largely stable over recent decades, but persistent decline in cold summers could desynchronize tree reproduction in the future. (A)** Observed synchrony within populations (purple, the average proportion of trees of the same population in the same reproductive state) and across populations (blue, the proportion of trees in the same state across all populations) has remained largely stable over recent decades. **(B)** In a hypothetical population, the synchrony is maintained while summers alternate between cold (around 17 °C) and warm (around 22 °C) conditions (green), but declines once the population experiences only warm summers (orange). X-axis shows hypothetical years. In **(A)** and **(B)**, we show median model predictions (solid lines) with 50% and 90% uncertainty intervals (shaded areas).

For population synchrony, in contrast to previous results, we found no clear evidence of declines in synchrony in recent decades, during a period of significant warming (Bogdziewicz *et al.*, 2020; Foest *et al.*, 2024). Synchrony across trees within a population varied over time but has not clearly changed in recent periods (recent declines in 2015 onward are well within the 90% range of previous periods, Fig. 5A). Similarly, synchrony between populations has been quite stable, from a previous mean of 84.3% (81.1 – 87.2%) before 2015 to one of 82.8% (78.9 – 85.2%) since 2015. This stability in synchrony across both scales suggests that masting in these beech populations may be more resilient to warming than previously predicted (Bogdziewicz *et al.*, 2021b).

Whether masting can continue to support forest regeneration in the next decades, however, may depend on how synchrony responds as the climate continues to warm. While we find no evidence for a positive feedback of summer warming that would trigger runaway seed production at the individual level, our results predict possible changes at the population level in synchrony as we lose the cold summers. Because cold summers act as potential hinge points for population synchronization, our results suggest that several consecutive years without cold summers decrease synchrony within a population (Fig. 5B). Determining the extent to which this 'breakdown' will happen in the coming decades, and producing robust spatiotemporal forecasts, will require better measurements of the temporal and spatial scales at which synchrony occurs (LaMontagne *et al.*, 2020). Our results show the value of building from individual-level seed production data, which is still incredibly rare, to understanding the population scale. Teasing out the full effects of warm summers versus spring frosts, however, will also require temporally finer resolution data than seed counts (e.g., flowering in Nussbaumer *et al.*, 2020)—ideally following buds from formation to flowering to fruits to measure losses at each stage.

Fundamentally synchrony in seed production is one piece of a larger puzzle. Understanding future forest regeneration requires disentangling the full sequence of reproductive and growth phenology stages that lead from a floral bud to an established tree. New data on bud formation, flowering, and seed production across years could improve models of masting. Improving our understanding of this is the first critical step of a much longer sequence could help pinpoint how climate change will ultimately impact forest regeneration. Climate change may lead to major shifts in the later stages—seed germination, seedling establishment and tree recruitment (Petrie *et al.*, 2023), with potential global impacts on forests and the services they provide. Our results highlight that we may fail to accurately forecast these shifts until we integrate the biological and climatic constraints that start with individual trees.

Methods

Tree-level reproduction and climate data

We used annual seed production data from 171 individual beech trees (*Fagus sylvatica*) across 12 populations in England, collected between 1980 and 2024, following the protocol described in previous publications (Packham *et al.*, 2008; Bogdziewicz *et al.*, 2021b; Hacket-Pain *et al.*, 2025). Populations contained between 8 and 21 trees and were monitored for 5 to 45 years. Within populations, individual time series varied in length (average length = 20.4 years) and contained some missing observations.

We extracted daily climatic data from ERA5-Land (Muñoz-Sabater *et al.*, 2021), at a 0.01 degree resolution, for the 12 sites. We computed three climatic variables for each year of observation: average maximum temperature in July and August (of the previous year), average spring temperature (April-May), and growing degree days (GDD) until the last frost day as a measure of spring frost risk (Vitasse & Rebetez, 2018).

A Hidden Markov model for individual tree reproduction

Structure of the model

We modeled individual tree seed production with a Hidden Markov Model (HMM), in a Bayesian framework using the probabilistic programming language Stan (Carpenter *et al.*, 2017). In this model, each tree k is assigned to a latent reproductive state $z_{k,i}$ each year i: a low reproductive state ($z_{k,i} = 1$) or a high reproductive state ($z_{k,i} = 2$). The latent state $z_{k,i}$ is conditionally dependent only on the state the preceding year (the *Markovian* assumption), such that for N years we have:

$$p(z_{k,1}, \dots, z_{k,N}) = p(z_{k,1}) \prod_{i=2}^{N} p(z_{k,i} \mid z_{k,i\text{-}1}) \tag{1}$$

The transitions between reproductive states are determined by a 2×2 matrix of probabilities:

$$\begin{bmatrix} p(z_{k,i} = 1 \mid z_{k,i\text{-}1} = 1) & p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1) \\ p(z_{k,i} = 1 \mid z_{k,i\text{-}1} = 2) & p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2) \end{bmatrix} \tag{2}$$

which is by definition equivalent to:

$$\begin{bmatrix} 1 - p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1) & p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1) \\ 1 - p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2) & p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2) \end{bmatrix} \tag{3}$$

Both the latent reproductive states and the transition probabilities were inferred from the seed data alone, i.e. the model identifies the most probable state sequence for each individual tree given its observed seed counts.

The latent state determines the observational model responsible for the observed seed count $y_{k,i}$. We modeled the seed production in a low reproductive state with a zero-inflated negative binomial distribution, and the one in a high reproductive state with a negative binomial distribution:

$$y_{k,i} \mid z_{k,i} = 1 \sim \begin{cases} 0 & \text{with probability } \theta \\ \text{NegBin}(\lambda_{\text{low}}, \psi_{\text{low}}) & \text{with probability } 1 - \theta \end{cases} \tag{4}$$

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{\text{high}}, \psi_{\text{high}}) \tag{5}$$

Influence of climate on state transition and seed production

We expected the climatic conditions experienced by a tree to influence both the transition matrix (Equation 3) and the seeds produced in a high reproductive state (Equation 5). Conversely, we assumed that climate does not impact seed production in a low reproductive state (Equation 4). In Equation 3, we assumed that only the transition from a low reproductive state to a high reproductive state was affected by climate (and thus, by construction, the probability of staying in a low reproductive as well):

$$p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1) = \text{logit}^{-1} \left(\alpha_{1\to2} + \beta^{\text{summer}}_{1\to2} \cdot (X^{\text{summer}}_{k,i\text{-}1} - X^{\text{summer}}_{\text{baseline}}) \right) \tag{6}$$

where $X_{k,i\text{-}1}^{\text{summer}}$ is the average maximum temperature during summer (the preceding year i-1), $\beta_{1\to2}^{summer}$ is the effect of these summer temperature, and $\text{logit}^{-1}(\alpha_{1\to2})$ is the transition probability when $X_{t,i\text{-}1}^{\text{summer}} = X_{\text{baseline}}^{\text{summer}}$.

We did not expect $X_{k,i\text{-}1}^{\text{summer}}$ to affect $p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2)$ because the temporal overlap between fruit development and floral bud initiation should prevent trees already in a high reproductive state from responding to favorable summer conditions (though we still checked that this effect was negligible, see Supp. Fig. S4 and S6).

In Equation 5, we assumed that both spring temperature $X_{k,i}^{\text{spring}}$ and spring frost risk $X_{k,i}^{\text{frost}}$ could impact the seed production in a high reproductive state, such that:

$$y_{k,i} \mid z_{k,i} = 2 \sim \text{NegBin}(\lambda_{k,i}^{\text{high}}, \psi_{\text{high}}) \tag{7}$$

$$\log\left(\lambda_{k,i}^{\text{high}}\right) = \log(\lambda_{\text{high}}) + \beta_{\text{high}}^{\text{spring}} \cdot (X_{k,i}^{\text{spring}} - X_{\text{baseline}}^{\text{spring}}) + \beta_{\text{high}}^{\text{frost}} \cdot (X_{k,i}^{\text{frost}} - X_{\text{baseline}}^{\text{frost}}) \tag{8}$$

Masting synchrony and model predictions

To quantify reproductive synchrony we aggregated individual tree states within and across populations, i.e. the proportion of trees sharing the same reproductive state (Fig. 5). Within populations, we computed synchrony as the maximum of the proportion of trees in a high or low reproductive state for each population, then averaged across populations. Across populations, we computed synchrony similarly, but we first averaged the proportions of trees in each state across all populations before taking the maximum (to avoid a population with more trees contributing more). Synchrony within populations can thus be higher than synchrony across populations when trees within a site tend to be in the same state but differ between sites. Since our model includes only two states, the minimum synchrony is 50% (i.e. at least 50% of the trees are in the same state).

To forecast the effects of summer warming (Fig. 4), we simulated seed production (and synchrony) for 50 trees representing an average population (from our data) with projected summer warming up to 7°C over one century (Schumacher *et al.*, 2024). To assess the role of biological constraints, we compared predictions from our model—in which reproductive constraints prevent trees from staying indefinitely in a high reproductive state—against a hypothetical model where summer temperatures had a similar effect on both $p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 1)$ (transition from low to high) and $p(z_{k,i} = 2 \mid z_{k,i\text{-}1} = 2)$ (persistence in high). To illustrate the role of cold summers in reproductive synchrony, we also simulated seed production for 50 trees first experiencing alternating cold (around 17 °C) and warm summers (around 22 °C), followed by a period of warm summers only (Fig. 5B).

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